Database Project Part 2:

Develop a logical data model reflecting the following requirements:

A. Derive relations (and FKs) from the conceptual model.

   Include the relational schema (e.g.: Staff(staffID, fName, lName, position, salary)).

Client -> Hire [1 .. 1 or many]
Hire -> Vehicle [1 or many .. 1]
Vehicle -> Site [1 or many .. 1]
Site -> Staff [1 .. 1 or many]

For 1:N relationships, The PK of the "1" side is placed onto the table of the "N" side.

- **clientNumber** (Client PK) is placed in the hire table.
- **siteNumber** (Site PK) is placed on the Staff table (already there).
- **vehicleRegistrationNumber** (Vehicle PK) is placed in the Hire table (already there).
- **siteNumber** (Site PK) is placed in the Vehicle table.

There are no M:N relationships, so no junction tables need to be created.

Foreign Keys are bold:

- Vehicles: registrationNumber, model, make, engineSize, capacity, currentMileage, dailyHireRate, curLocation, **siteNumber**
- -Sites: siteNumber, phoneNumber, faxNumber, address
- Clients: clientNumber, name, address, phoneNumber, DOB, licenseNumber
- Staff: staffNumber, name, homeAddress, phoneNumber, DOB, sex, startDate, jobTitle, salary, **siteNumber**
- Hire Agreements: hireNumber, **clientNumber**, clientName, clientAddress, clientPhoneNumber, hireStartDate, hireTerminationDate, **vehicleRegistrationNumber**, vehicleModel, vehicleMake, startMileage, endMileage

B. Validate the logical model using normalization to 3NF.

   The logical model is already in 1st normal form, as each cell would contain just one value and each column in a table has a unique name.

   For the logical model to be in 2NF, each non-key attribute must depend wholly on the primary key. The primary keys are:

   - Vehicles: **<u>registrationNumber</u>**
   - Sites: **<u>siteNumber</u>**
   - Clients: **<u>clientNumber</u>**

- Staff: **staffNumber**
- Hire Agreements: **hireNumber**

Ignoring foreign keys, the Vehicles, Sites, Clients, Staff tables are in 2NF. However, the Hire Agreements table is not. The attributes clientName, clientAddress, clientPhoneNumber, vehicleModel, and vehicleMake are not dependent on the primary key. Leaving them in could cause update anomalies. Having the foreign keys to the client Clients and Vehicles tables is enough. The new Hire Agreements table looks like:

- Hire Agreements: hireNumber, **clientNumber**, hireStartDate, hireTerminationDate, **vehicleRegistrationNumber**, startMileage, endMileage

To ensure that the table is in 3NF, all transitive dependencies should be removed.

In the Vehicles table, make and model form a transitive dependency. The make of a vehicle does not depend on its registration number. This can be fixed by taking out make from the table and creating a new table called model:

- Models: model, make

With the assumption that a make is tied to a single model.

engineSize and capacity are not transitive dependencies. It is possible for a car to have custom modifications or seating arrangements.

If it is assumed that all job titles make the same salary, jobTitle and salary could form a transitive dependency in the Staff table. However , I believe it is a safe assumption to assume that the same position could have different salaries. For example, a long-time employee in a position could have their pay raised.

Outside of these cases, the logical model is in 3NF. The final model is:

- Vehicles: **registrationNumber**, model, engineSize, capacity, currentMileage, dailyHireRate, curLocation, **siteNumber**
- Models: **model**, make
- Sites: **siteNumber**, phoneNumber, faxNumber, address
- Clients: **clientNumber**, name, address, phoneNumber, DOB, licenseNumber
- Staff: **staffNumber**, name, homeAddress, phoneNumber, DOB, sex, startDate, jobTitle, salary, **siteNumber**
- Hire Agreements: **hireNumber**, **clientNumber**, hireStartDate, hireTerminationDate, **vehicleRegistrationNumber**, startMileage, endMileage

Primary Key **Foreign Key**

C. Validate the logical model against 5 user transactions. (Note: These will be then implemented in 3c.)
    a. You must include the query (question) in plain English.

b. The majority of queries (questions) should involve more than one entity.
c. You must explain verbally what steps you follow using the relationships between the entities (no code!).

Transaction 1:

a. Query: Which staff members work at the site where a specific vehicle is located?
b. Entities Involved: Vehicle and Staff
c. Steps: Locate the specific vehicle in the *Vehicle* entity using its *registrationNumber* to find its associated *siteNumber*. Next, take that *siteNumber* and use it to search the *Staff* entity, returning the names of all staff members who have that matching *siteNumber*.

Transaction 2:

a. Query: Which staff members work at the site where a specific vehicle is located?
b. Entities Involved: Vehicle and Staff
c. Steps: Query the *HireAgreement* entity using the *registrationNumber* and *hireStartDate* to find the associated *clientNumber*. Then, take that *clientNumber* to search the *Client* entity and retrieve the client's *address*.

Transaction 3:

a. Query: How many vehicles are currently allocated to the site where Sebatian works?
b. Entities Involved: Staff and Vehicle
c. Search the *Staff* entity using the *name* "Sebatian" to retrieve his *siteNumber*. Then, use that *siteNumber* to query the *Vehicle* entity and count the total number of vehicles that share that *siteNumber*.

Transaction 4:

a. Query: What are the vehicle details (make and model) for all active hire agreements signed by a specific client?
b. Entities Involved: HireAgreement, Vehicle, and Models
c. Search the *HireAgreement* entity using the *clientNumber* to find all associated *vehicleRegistrationNumber* records for that client. Then, take those registration numbers and use them to query the *Vehicle* entity to find the matching *model*. Finally, use that *model* to query the *Models* entity, returning the *make* for each match.

Transaction 5:

a. Query: What is the phone number of the site where a specific client's rented vehicle was originally allocated?
b. Entities Involved: HireAgreement, Vehicle, and Site

c. Query the HireAgreement entity using the *cilentNumber* to find the *registrationNumber*. Use that registration number to query the *Vehicle* entity to find its siteNumber. Finally, use that site number to query the Site entity and retrieve the *phoneNumber*.

D. Define integrity constraints:
   a. Primary key constraints.
      i. Site: *siteNumber* must be unique and cannot be NULL
      ii. Models: *model* must be unique and cannot be NULL
      iii. Vehicle: *registrationNumber* must be unique and cannot be NULL
      iv. Client: *clientNumber* must be unique and cannot be NULL
      v. Staff: *staffNumber* must be unique and cannot be NULL
      vi. Hire Agreement: *hireNumber* must be unique and cannot be NULL
   b. Referential integrity/Foreign key constraints.
      i. The *siteNumber* in the Vehicle relation must match an existing *siteNumber* in the Site relation, or be NULL
      ii. The *model* in the Vehicle table must match an existing *model* in the Models table
      iii. The *siteNumber* in the Staff relation must match an existing *siteNumber* in the Site relation.
      iv. The *clientNumber* in the HireAgreement relation must match an existing *clientNumber* in the Client relation.
      v. The *registrationNumber* in the HireAgreement relation must match an existing *registrationNumber* in the Vehicle relation
   c. Alternate key constraints (if any).
      i. Client: *licenseNumber* must be unique. Even though *clientNumber* is the primary key, two clients cannot legally hold the exact same driver's license number.
   d. Required data.
      i. Client: *name, DOB, phoneNumber, and licenseNumber*
      ii. Models: *make* must be NOT NULL.
      iii. Vehicle: *dailyHireRate* must be NOT NULL
      iv. Staff: *name, startDate, and salary* must be NOT NULL
      v. Hire Agreement: *hireStartDate, startMileage, clientNumber,* and *vehicleRegistrationNumber* must be NOT NULL
   e. Attribute domain constraints.
      i. *sex* in the Staff relation must be restricted to specific values (M, F)
      ii. *dailyHireRate* in the Vehicle relation and *salary* in the Staff relation must be positive numeric values greater than 0

iii. *currentMileage* in the Vehicle relation, and *startMileage/endMileage* in the Hire Agreement relation must be positive integers (>= 0).

iv. All date fields (*DOB, startDate, hireStartDate, hireTerminationDate*) must be formatted as valid calendar dates.

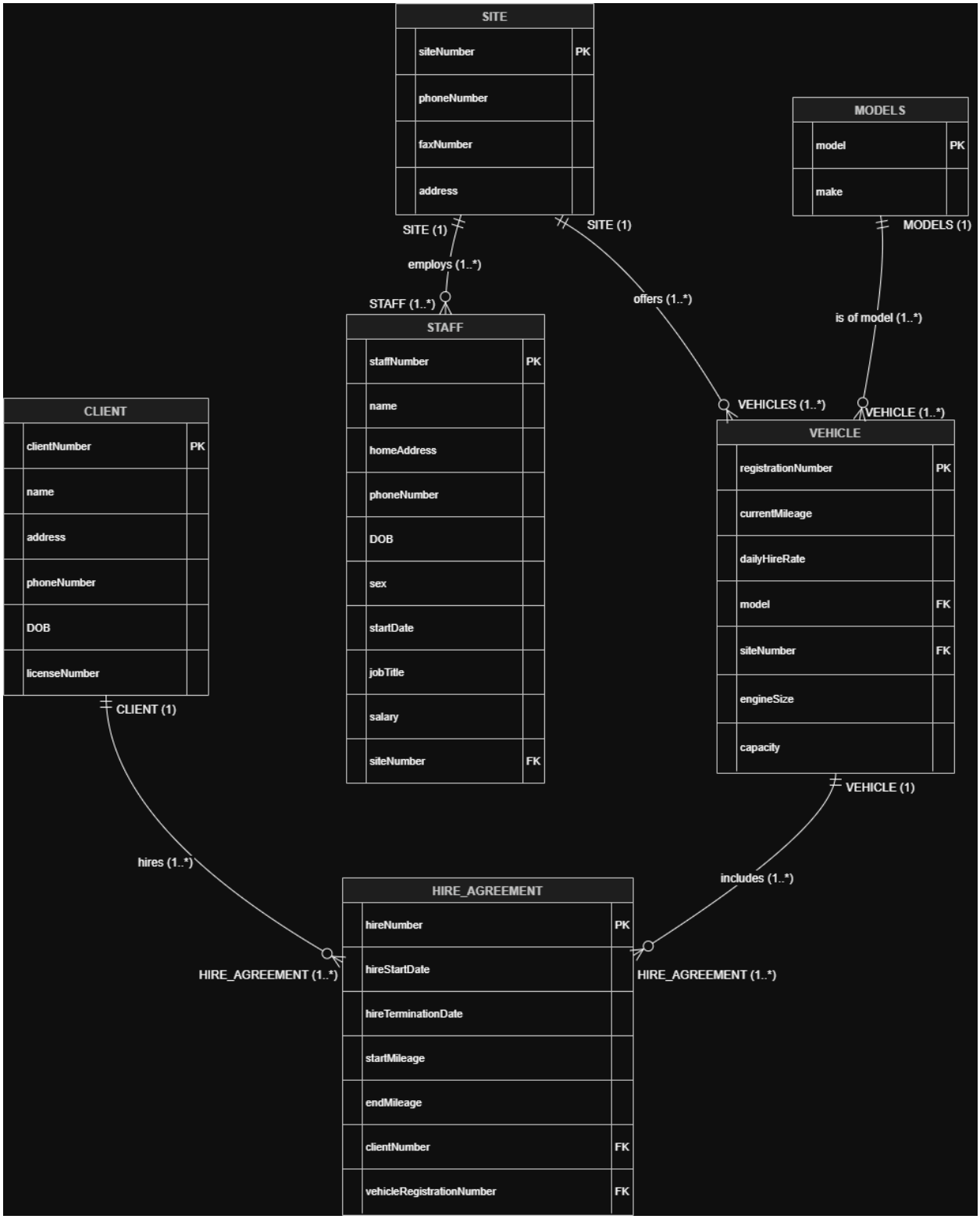
f. General constraints (if any).

i. In the Hire Agreement, the *hireTerminationDate* must be greater than or equal to the *hireStartDate.*

ii. In the Hire Agreement, the *endMileage* must be greater than or equal to the *startMileage.*

iii. In the Staff and Client tables, the *DOB* must ensure the individual is of legal working or driving age (e.g., >= 18 years old) at the time of their start date or hire start date.

E. Generate the E-R diagram for the logical level (contains FKs as attributes).



Notes: *A report must be created including detailed documentation of each of the steps addressing all the required items. ER diagrams for the logical model must be included in the report. All assumptions made in the design must be clearly stated.*

Model Assumptions:

1. Each model belongs to exactly one make.
2. A make can have many models.
3. All vehicles of the same model share the same engine size and capacity.
4. Each vehicle has a unique registration number regardless of model or make.